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SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

PREDICTIVE ANALYSIS

For

Bachelor of Technology

In

Computer Science and Engineering

**Predicting Online Chess Game Outcomes
from Player Ratings, Time Controls, and Openings Using Machine Learning**

Continuous Assessment

Predictive Analysis Comprehensive Project Report

Submitted by

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ABSTRACT

This project investigates how much of the outcome of an online chess game can be predicted before play begins. Using a public Kaggle sample of 20,058 Lichess games, four classifiers were trained to predict white win, black win, or draw from player ratings, the rating gap, time control, rated status, and opening information. The workflow follows the Knowledge Discovery in Databases (KDD) process from data selection and preprocessing through transformation, modelling, evaluation, and interpretation.

After duplicate removal, 19,113 games remained. The analysis deliberately excludes post-game variables such as number of turns, victory status, moves, and last-move time so that the reported performance reflects information genuinely available before a game begins. Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting were compared against majority-class and higher-rated-player baselines.

The models achieve approximately 62–63% accuracy, with Gradient Boosting reaching 0.628 accuracy and Logistic Regression providing the strongest interpretability and competitive log-loss. The rating gap and Elo expected score account for most of the predictive signal, while opening and time-control effects are statistically detectable but small in practical terms. The draw class remains difficult to predict and the reference model records an F1 score of 0.00 for draws.

A leakage experiment demonstrates why post-game variables must be excluded: adding number of turns and victory status raises test accuracy to about 90.3%, but this information is unavailable at prediction time. The project therefore treats the 62–63% range as the meaningful result. The findings show that, for this dataset, information quality and availability constrain predictive performance more strongly than the choice among the tested algorithms.

Keywords: Predictive Analysis, Machine Learning, Chess Outcome Prediction, KDD, Logistic Regression, Random Forest, Gradient Boosting, Data Leakage



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CONTENTS

Sr. No.	Topic	Page No.
1	INTRODUCTION & PROBLEM STATEMENT	6
	1.1 Background	6
	1.2 Problem Statement	6
	1.3 Objectives	6
	1.4 Scope and Boundaries	6
2	LITERATURE REVIEW	7
	2.1 Rating Systems as Predictors	7
	2.2 The KDD Process and Predictive Modelling	7
	2.3 Classifiers Used	7
	2.4 Data Leakage	7
	2.5 Gap Addressed	7
3	PREDICTIVE MODELLING METHODOLOGY	8
	3.1 Data Selection and Understanding	8
	3.2 Preprocessing	8
	3.3 Transformation and Feature Engineering	8
	3.4 Exploratory Analysis and Feature Selection	9
	3.5 Model Building	10
	3.6 Evaluation Plan	10
4	RESULTS & DISCUSSION	11
	4.1 What the Data Looks Like	11
	4.2 Do Openings and Time Controls Matter?	11
	4.3 Model Performance	12
	4.4 Where the Model Goes Wrong	12
	4.5 Which Factors Drive the Predictions	14
	4.6 Why This Is Not a Perfect Model: Leakage Check	14
	4.7 Discussion and Limitations	14
5	CONCLUSION	15
	REFERENCES	16



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TABLE OF FIGURES

Fig No.	Figure Name	Page No.
1	Outcome distribution, outcome share by rating gap, and rating gap by outcome	9
2	Outcome shares by opening family, time category and rated status	10
3	Accuracy, macro F1 and ROC-AUC for each model on the held-out test set	12
4	Confusion matrix of the Logistic Regression reference model	13
5	Permutation importance for the Logistic Regression reference model	14



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TABLE OF TABLES

Table No.	Table Name	Page No.
1	Data-cleaning checks and actions	8
2	White win rate by rating gap	11
3	Selected openings with at least 100 games	11
4	Test-set performance of baselines and classifiers	12

CHAPTER 1

INTRODUCTION & PROBLEM STATEMENT

1.1 Background

Online chess platforms match millions of games a day. Players are paired using a rating, a number that is supposed to say how strong someone is, and the whole idea of fair matchmaking rests on that number meaning something. If a 1500 player is paired with a 1700 player, how likely is the stronger player to win? Does it matter if the game is a one-minute bullet game or a fifteen-minute rapid game? Do some openings quietly favour white or black?

These questions suit predictive analytics well. The inputs are measurable before play starts, the outcome has three clear categories, and a real platform produces the data at scale. Chess is also a good teaching case because it contains genuine uncertainty: even a large rating gap does not settle the result.

1.2 Problem Statement

To develop a predictive analytics model that forecasts the outcome of an online chess game (white win, black win or draw) using only information available before or at the start of play: white and black ratings, the rating gap, the time control, whether the game is rated, and the opening played.

1.3 Objectives

- Apply the KDD process end to end on a real dataset, from selection to interpretation.
- Build and compare four classifiers for three-class outcome prediction, against sensible baselines.
- Find out which pre-game factors (rating gap, opening, time control) really influence the result.
- Avoid data leakage, and show what leakage does to the numbers.

1.4 Scope and Boundaries

The project uses a public Kaggle sample of 20,058 Lichess games. It does not look at the moves themselves, at engine evaluations, or at player history, because none of those are available before a game begins. The practical uses considered are checking matchmaking quality, coaching around opening choice, and understanding upset probabilities.

CHAPTER 2

LITERATURE REVIEW

2.1 Rating Systems as Predictors

The modern idea of predicting a chess result from ratings goes back to Elo (1978), who modelled a player's performance as a random variable and defined the expected score of a match from the rating difference alone. In that model a 200 point advantage gives the stronger player a bit over a 75% expected score. Lichess does not use Elo directly; it uses Glicko-2, an extension by Glickman that adds a rating deviation and a volatility term so the system knows how confident it is about each rating (Glickman, 2001). Both systems are built to produce probabilities, which is why this project includes the classical Elo win expectancy as an engineered feature and why a simple rating rule makes such a hard baseline to beat.

2.2 The KDD Process and Predictive Modelling

Fayyad, Piatetsky-Shapiro and Smyth (1996) described knowledge discovery as a sequence: selecting data, preprocessing it, transforming it, mining patterns, and then interpreting what was found. The course follows this framing and so does this project. The point of the framework is that the model is only one step; many problems come from decisions made earlier, such as what data to include or how to treat missing values and duplicates.

2.3 Classifiers Used

Logistic regression is the standard interpretable baseline for classification and gives calibrated probabilities when the features are well behaved. Decision trees offer rule-like explanations but overfit easily. Random forests (Breiman, 2001) average many trees and cope well with non-linear effects and mixed feature types. Gradient boosting builds trees sequentially, each one correcting the last, and is often the strongest tabular method in practice. The project uses the scikit-learn implementations (Pedregosa et al., 2011).

2.4 Data Leakage

Kaufman, Rosset, Perlich and Stitelman (2012) describe leakage as the introduction of information into the modelling process that would not be legitimately available at prediction time, and note that it is one of the most common reasons a model looks excellent in testing and fails in use. In this dataset the danger is obvious: the number of turns and the victory status are recorded after the game. They are therefore treated as forbidden inputs and their effect is demonstrated in Section 4.6.

2.5 Gap Addressed

Many quick analyses of this dataset report very high accuracy because they let the post-game columns in. This project restricts itself to pre-game information, compares against honest baselines, and explains what the results can and cannot support.

CHAPTER 3

PREDICTIVE MODELLING METHODOLOGY

The workflow follows the KDD stages. Everything below is implemented in the accompanying Jupyter notebook, which is executed and saved with its outputs.

3.1 Data Selection and Understanding

The dataset has 20,058 games and 16 columns (Kaggle, Chess Game Dataset by J. Mitchell). The target is winner. The inputs kept are the two ratings, the rated flag, the time control code (for example, “15+2” meaning 15 minutes plus a 2 second increment), the ECO opening code and the opening length in half-moves. The project removes turns, victory_status, moves and last_move_at as post-game information. It also drops created_at, because it is stored in truncated scientific notation (1.50421E+12) and therefore carries almost no information, and the id columns.

3.2 Preprocessing

Check	What I found	Action
Duplicate game IDs	945 repeated IDs	Removed, leaving 19,113 games
Missing values	None in the file	None needed
Rating outliers (IQR rule)	A small number of very high or low ratings	Kept: they are real players, not errors
Invalid ratings (zero or below)	None	Filter kept as a safeguard

Table 1. Data-cleaning checks and actions.

3.3 Transformation and Feature Engineering

- Rating features: rating difference (white minus black), its absolute value, the average rating, and the Elo expected score for white.
- Time features: the time code is split into base minutes and increment seconds, then grouped into Bullet, Blitz, Rapid and Classical using an estimated game length of base time plus 40 moves of increment.
- Opening features: ECO family (A to E) and the 30 most common ECO codes, with everything rarer placed in an “Other” group so the model is not drowned in sparse categories.
- Encoding: numeric features are standardised and categorical features one-hot encoded inside a scikit-learn pipeline, so the same steps are applied to training and test data without leakage.

3.4 Exploratory Analysis and Feature Selection

The most striking pattern is how cleanly the white win rate rises with the rating gap. Opening family and time category show smaller differences, and chi-square tests confirm the associations are statistically real (Section 4.2) even though they are small in practical terms. A correlation heatmap and permutation importance were used to check redundancy and relative influence. Several rating features are naturally correlated, which is fine for tree models and handled by regularisation in the logistic regression.

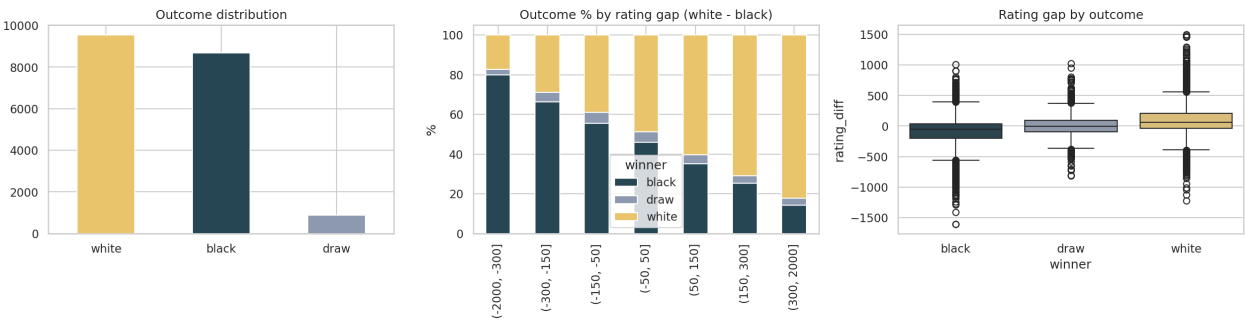


Fig 1: Outcome distribution, outcome share by rating gap, and rating gap by outcome.

3.5 Model Building

The data were split 80/20 with stratification on the outcome, so the test set (3,823 games) keeps the same class mix as the training set. Logistic Regression, a Decision Tree, a Random Forest and Gradient Boosting were trained. Hyper-parameters were tuned by five-fold cross-validation on the training data only, optimising log-loss so that the models are rewarded for good probabilities and not just correct labels. The test set was touched once, at the end.

3.6 Evaluation Plan

Because draws are only 4.6% of games, accuracy alone is misleading. The project reports accuracy, balanced accuracy, macro precision, recall and F1, log-loss and one-vs-rest ROC-AUC, plus confusion matrices and five-fold cross-validation on the full data. Two baselines frame the results: always predict the majority class, and “the higher-rated player wins”.

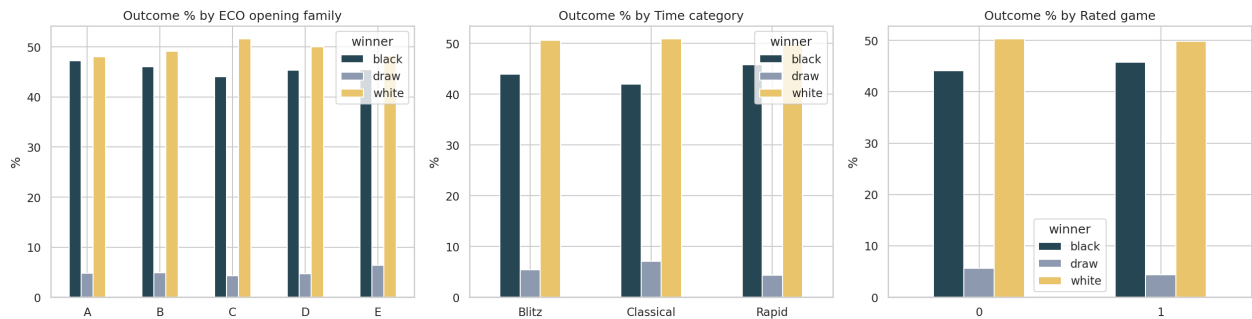


Fig 2: Outcome shares by opening family, time category and rated status.

CHAPTER 4

RESULTS & DISCUSSION

4.1 What the Data Looks Like

After cleaning, the 19,113 games split into 9,545 white wins (49.9%), 8,680 black wins (45.4%) and 888 draws (4.6%). White enjoys a small first-move advantage. The effect of the rating gap is far bigger.

White minus black rating	White win rate
(-inf, -300]	17.2%
(-300, -150]	28.8%
(-150, -50]	39.1%
(-50, 50]	48.7%
(50, 150]	60.2%
(150, 300]	70.8%
(300, inf)	82.1%

Table 2. White win rate by rating gap.

When white is more than 300 points weaker, white wins about one game in six; when white is more than 300 points stronger, white wins more than four in five. Among games between near-equals, the result is close to a coin flip.

4.2 Do Openings and Time Controls Matter?

Chi-square tests show that the outcome is significantly associated with the opening group (chi-square 243.4 on 60 degrees of freedom, $p < 0.001$), the time category (35.3 on 4, $p < 0.001$), the rated flag (10.6 on 2, $p = 0.005$) and the ECO family (19.4 on 8, $p = 0.013$). So these variables are not noise. But a significant association is not the same as a useful predictor, and the next sections show how little they add.

ECO	Most common name in the data	Games	White win	Black win
B40	Sicilian Defense	120	39.2%	52.5%
B20	Sicilian Defense	551	39.4%	56.4%
A00	Van't Kruijs Opening	948	39.5%	56.8%
B00	Owen Defense	574	59.8%	35.7%
C24	Bishop's Opening	108	62.0%	33.3%
D06	Queen's Gambit Refused	167	63.5%	34.7%

Table 3. Selected openings with at least 100 games.

Among openings with at least 100 games, the listed openings show the widest swings. Even the most lopsided ones are separated from 50% by only a few points, and these are raw rates that also mix in who happens to be playing them.

4.3 Model Performance

Model	Accuracy	Balanced acc.	Macro F1	Log-loss	ROC-AUC
Majority class (baseline)	0.499	0.333	0.222	18.05	0.500
Higher rating wins (baseline)	0.622	0.435	0.424	n/a	n/a
Logistic Regression	0.624	0.435	0.424	0.7627	0.688
Decision Tree	0.620	0.429	0.417	0.7691	0.659
Random Forest	0.623	0.434	0.423	0.7627	0.685
Gradient Boosting	0.628	0.437	0.427	0.7632	0.684

Table 4. Test-set performance (3,823 games).

Every model beats the 50% baseline by about twelve points, but none of them beats the plain rating rule by more than half a point. Gradient Boosting has the highest accuracy and Logistic Regression and Random Forest the best log-loss, yet five-fold cross-validated accuracy for all four lies between 61.9% and 62.2% with a standard deviation of about one point. In other words, the differences between the models are smaller than the noise. Logistic Regression is used as the reference model because it matches the others, is the simplest, and gives the best-calibrated probabilities.

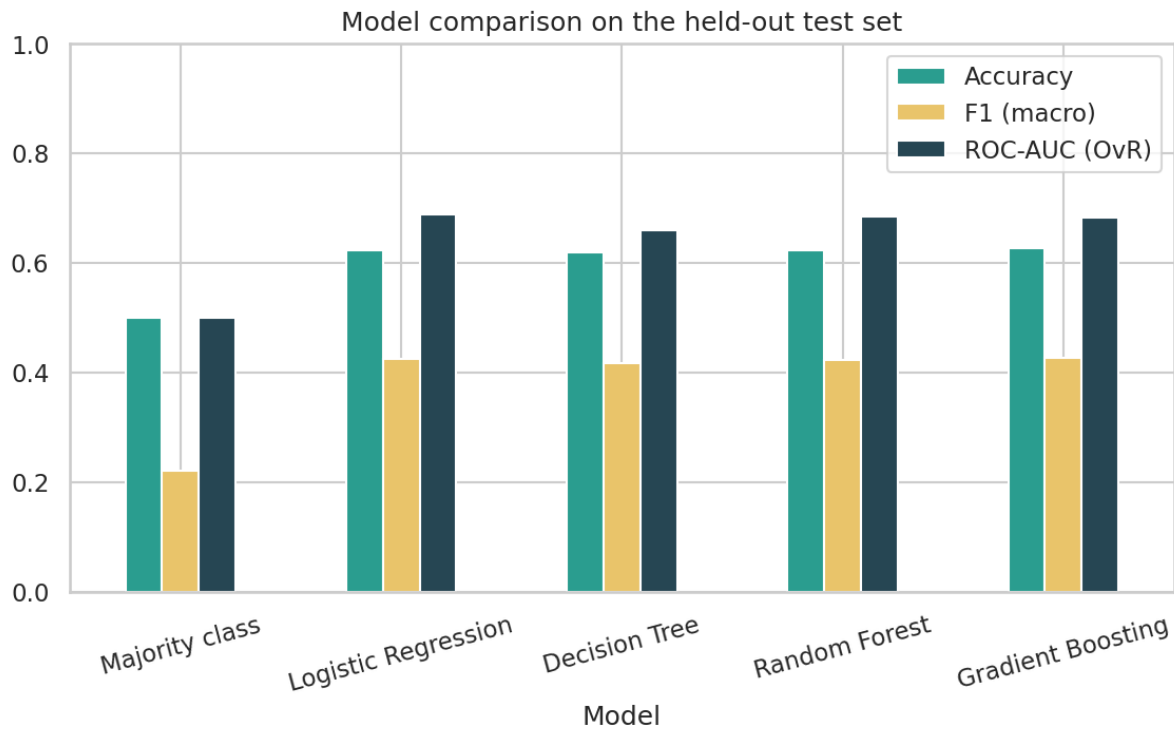


Fig 3: Accuracy, macro F1 and ROC-AUC for each model on the held-out test set.

4.4 Where the Model Goes Wrong

Per-class F1 for the reference model is 0.66 for white wins and 0.61 for black wins, but 0.00 for draws. The model never predicts a draw, because no combination of pre-game inputs makes a draw the single most likely result. Drawn games are rare and tend to come from long, balanced fights that pre-game data cannot describe. This is why macro F1 sits at about 0.42: two classes are handled reasonably and one is not handled at all.



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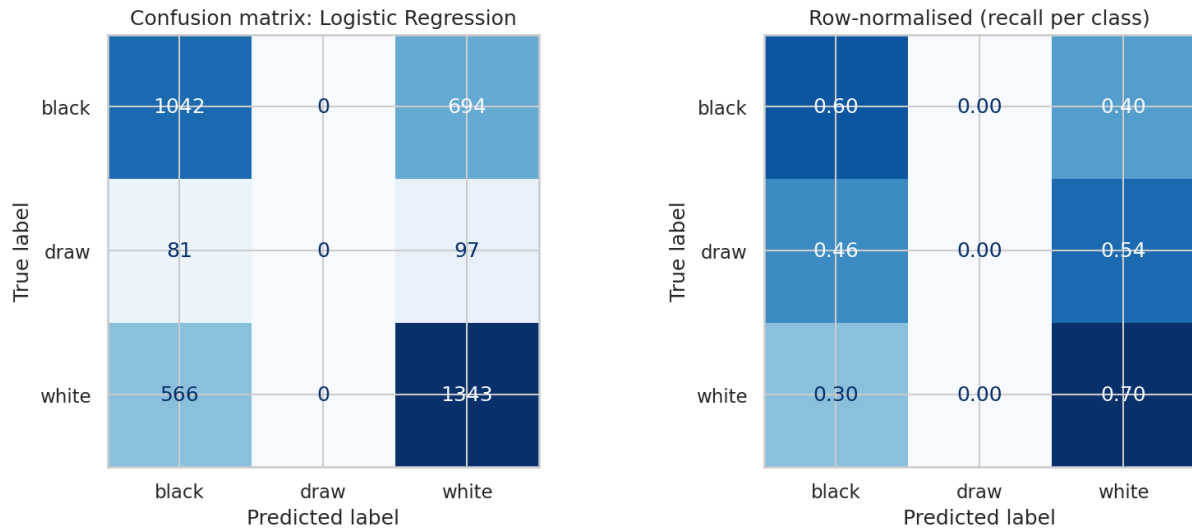


Fig 4: Confusion matrix of the reference model (counts and row-normalised recall).

4.5 Which Factors Drive the Predictions

Permutation importance, which measures how much the model gets worse when a feature is shuffled, tells the same story as the exploratory analysis. The Elo expected score is far ahead of everything else, followed by the rating difference and the individual ratings. The opening group, the time category and the rated flag contribute very little, each shifting log-loss by only a few thousandths.

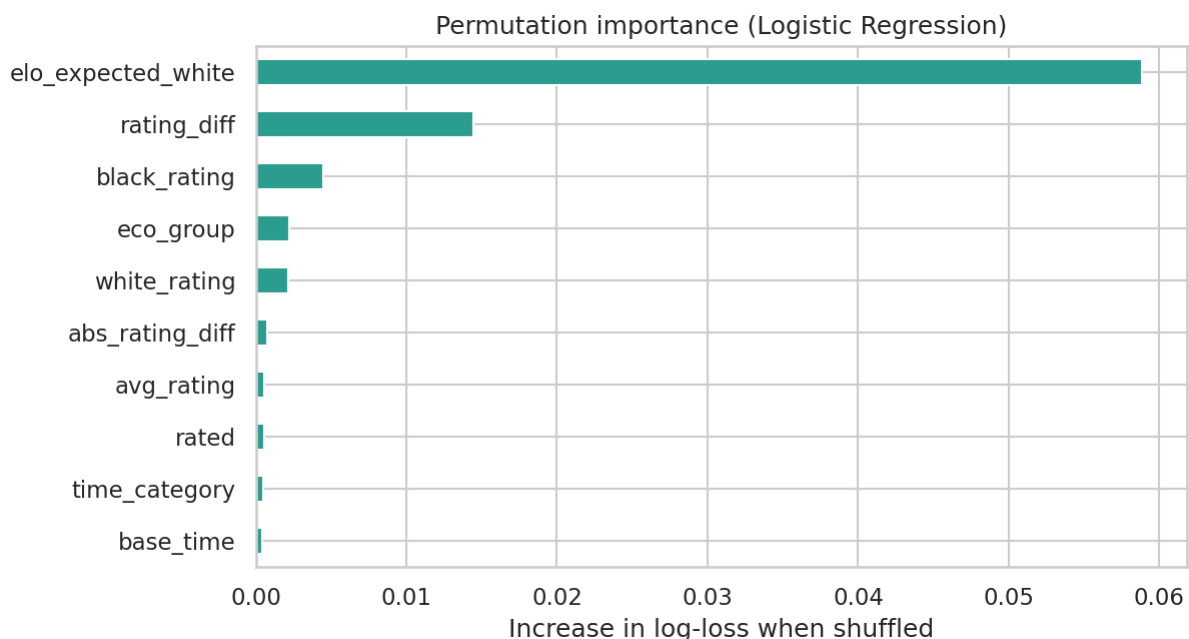


Fig 5: Permutation importance for the reference model (increase in log-loss when a feature is shuffled).

4.6 Why This Is Not a Perfect Model: A Leakage Check

To see what leakage does, the project added the number of turns and the victory status to an otherwise similar gradient boosting model. Test accuracy rose from 62% to 90.3%. This looks like a breakthrough and is not: if a game lasts an odd number of turns, white made the last move, and a game that ends in checkmate or resignation tells you who was losing. All of that is known only after the game. Such a model could never be used to predict a game that has not been played, so the 62 to 63% range is treated as the real result and the 90% figure as a warning.

4.7 Discussion and Limitations

- The ceiling is set by the information, not the algorithm. Four very different models land within half a point of each other and of a one-line rule.
- Practical use: the model is a good probability estimator for matchmaking and for expectation-setting (“you have about a 30% chance in this pairing”), but it is not a game-by-game oracle.
- Opening and time-control effects are real but small. They would matter more if player strength were controlled or if the opening were analysed at the position level.
- Limitations: a single platform and period, about 19,000 games, many players appearing only once (so per-player history is unusable), and no engine evaluation.
- Not done: SMOTE or other resampling. With a 4.6% draw class and no pre-game signal for it, resampling mostly produces more false draws, so no gain from it is claimed.

CHAPTER 5

CONCLUSION

The KDD process was applied to predict online chess outcomes using only pre-game information. After removing duplicates and leakage columns, engineering rating, time and opening features, and tuning four classifiers with cross-validation, the best models reach about 62 to 63% accuracy and a ROC-AUC near 0.69. That is twelve points above guessing and roughly equal to the rule that the higher-rated player wins.

The rating gap explains almost everything that can be explained. Openings and time controls show statistically significant but practically small effects, and draws are not predictable before the game. The leakage experiment shows how easily this dataset produces a misleadingly high score, which is why the cautious number is the more useful one.

If the project were continued, the first step would be to obtain a larger Lichess export with player history, so that recent form and head-to-head records could be added. After that, opening positions could be evaluated with a chess engine to test whether a position-level score adds signal beyond the ECO code. Neither is available in this dataset, so they remain future work.



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